Statistics for Computer Scientists

Lecture 8: Comparing two groups

7 November 2025

Universiteit Leiden
The Netherlands

Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
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14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
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This lecture

- Recap lecture 7 (statistical testing)

- Dependent vs independent samples (*paired* vs *unpaired*)
- Comparing means in two groups
  - Confidence interval
  - Significance test
- Comparing proportions in two groups
- Bootstrap sampling

Recap statistical testing

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Of how many parts does a statistical test consist?

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Statistical testing is useful when you want to

1. describe a sample
2. estimate a parameter
3. generate a hypothesis
4. assess a hypothesis

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Compared to a confidence interval, a statistical test is

1. more informative
2. equally informative
3. less informative

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A p-value represents

1. the likeliness of the observed test statistic if the null hypothesis is true
2. the likeliness of the observed test statistic if the null hypothesis is false
3. the likeliness of the observed test statistic if the alternative hypothesis is true
4. the likeliness of the observed test statistic if the alternative hypothesis is false

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A p-value does NOT become smaller when

1. the sample size increases
2. the effect size decreases
3. the confidence level (alpha) increases
4. the standard error decreases
5. the test statistic increases

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Significance tests

- **Hypothesis**: a prediction about a population, expressed in terms of parameters for given variables

- **Significance test**: uses data to summarize evidence about a hypothesis by comparing sample estimates of parameters to values predicted by the hypothesis

- We answer questions such as: "if the (null) hypothesis were true, how (un)likely would it be to get the estimate we obtained?"

Five parts

1. Assumptions
2. Hypotheses
3. Test statistic
4. P-value
5. Conclusion

Two tests covered so far

Univariate analysis

- t-test for a single mean: H_0: $\mu = \mu_0$
  - "Is the average height of Dutch students equal to 175 cm?"
- z-test for a single proportion: H_0: $\pi = \pi_0$
  - "Is the proportion of people with cats in this room equal to 0.5?"
- One-sided and two-sided alternatives

Comparing two groups
Dependent vs independent

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Two groups

Univariate analysis

- t-test for a single sample:

  is the mean equal to a certain number?

- t-test for two groups/samples:

  is the mean in both groups the same?
  (or: is the difference in means of the groups 0?)

Bivariate analysis

Bivariate analysis

- Univariate = one variable
- Bivariate = two variables
- Multivariate = multiple variables (typically > 2)

- Bivariate setting
  - **Explanatory** variable: defines the groups being compared (or more generally, the "x-axis")
  - **Response** variable: the variable on which comparisons are made

Dependent vs independent

- Dependent samples:
  - Same subjects in both groups, e.g., with longitudinal studies
    ("is the mean finishing time of people's second marathon lower than their first?")
  - Or related subjects in both groups
    ("is the mean happiness score of partners in a relationship the same?")
  - In general, can map (or *pair*) each observation in group 1 to one observation in group 2

- Independent samples:
  - The two groups are selected independently of each other

Dependent vs independent

- Variability in difference scores is lower with dependent samples
- Elimination of other sources of variability, more controlled

- Example: mean finishing times of first and second marathon
- Dependent samples: same people in both groups
- Independent samples: different people in both groups
  - Difference in finishing times can be due to other factors! Maybe one group is younger, fitter, etc…

- So: differences in calculation of standard error!

Comparing means in two groups

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Comparing two means

- Inference about $\mu_2 - \mu_1$ (usually: is it 0?)
- Parameter $\mu_2 - \mu_1$, estimated by statistic $\bar{y}_2 - \bar{y}_1$

- Compare with confidence interval and/or significance test

- Interval approach: make CI about $\mu_2 - \mu_1$
  - If 0 is not included in the interval, we are 95% (or 99%, etc) certain that the means in both populations are not the same

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CI for difference of two means (unpaired)

- For large random samples (*or for small random samples from normal population distributions*), the **sampling distribution** of $(\mu_2 - \mu_1)$ is approximately normal

- Remember CI for one mean: $\bar{y} \pm t * se$, where $se = \frac{s}{\sqrt{n}} = \sqrt{\frac{s^2}{n}}$

- CI for difference of means: $(\bar{y}_2 - \bar{y}_1) \pm t * se$, where $se = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
- t-score needed: how many df?

Degrees of freedom

- Remember for a single mean: $df = n - 1$

- More complicated when there are two groups

- If both n_1 and n_2 are > 30: interpret t-score as z-score (it's still a t-score though!)
- Else, if $s_1 = s_2$ and $n_1 = n_2$: $df = (n_1 - 1) + (n_2 - 1) = n_1 + n_2 - 2$
- Otherwise: use software

CI for two means (unpaired) - example

- "Do men and women have the same number of close friends?"
- Explanatory variable is sex, response variable is number of friends

- 486 women, $\bar{y}_F = 8.3$, $s_F = 15.6$
- 354 men, $\bar{y}_M = 8.9$, $s_M = 15.5$

- $(\bar{y}_2 - \bar{y}_1) \pm t * se$, where $se = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Example

486 women, $\bar{y}_F = 8.3$, $s_F = 15.6$

354 men, $\bar{y}_M = 8.9$, $s_M = 15.5$

- $se = \sqrt{\frac{s_F^2}{n_F} + \frac{s_M^2}{n_M}} = \sqrt{\frac{15.6^2}{486} + \frac{15.5^2}{354}} = 1.09$

- 95% CI: $(\bar{y}_F - \bar{y}_M) \pm t * se = (8.3 - 8.9) \pm 1.96 * 1.09 = -0.6 \pm 2.1 = (-1.5, 2.7)$
- Or the other way around $(\bar{y}_M - \bar{y}_F)$, gives $(-2.7, 1.5)$

- Interpretation?

Example

- 95% CI: (-1.5, 2.7)

- We can be 95% confident that the population mean number of close friends for women is between 1.5 fewer and 2.7 more than for men.
- 0 is in interval, so “no difference”

- Note: these population distributions are not normal!
  But because large *n*, CLT applies

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Significance test for two means

Instead of CI, we can also test whether two means are the same

1. **Assumptions**

Same as for t-test about single mean, but for each of two groups

2. **Hypotheses**

$H_0: \mu_1 = \mu_2 \quad (\mu_2 - \mu_1 = 0)$

$H_a: \mu_1 \neq \mu_2 \quad (\mu_2 - \mu_1 \neq 0)$

(or one-sided alternative)

Significance test for two means

3. Test statistic

As usual, t is the number of se between the estimate and the value under H_0

$t = \frac{(\bar{y}_2 - \bar{y}_1) - 0}{se}$, where $se = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ (same formula as for CI)

4. P-value & 5. Conclusion

Same as for t-test about single mean; read p from Table B, and don't forget to relate back to the context

Sig. test for two means - same example

- "Do men and women have the same number of close friends?"
- Explanatory variable is sex, response variable is number of friends

- 486 women, $\bar{y}_F = 8.3$, $s_F = 15.6$
- 354 men, $\bar{y}_M = 8.9$, $s_M = 15.5$

Example

1. **Assumptions**

   - Data are obtained using randomization
   - The variable of interest is quantitative
   - Population distribution is normal (we have both n > 30, so still OK)

2. **Hypotheses**

   $H_0: \mu_F - \mu_M = 0$

   $H_a: \mu_F - \mu_M \neq 0$

Example

3. Test statistic

486 women, $\bar{y}_F = 8.3$, $s_F = 15.6$ 354 men, $\bar{y}_M = 8.9$, $s_M = 15.5$

$$\text{se} = \sqrt{\frac{s_F^2}{n_F} + \frac{s_M^2}{n_M}} = \sqrt{\frac{15.6^2}{486} + \frac{15.5^2}{354}} = 1.09$$ (same as for CI)

$$t = \frac{(\bar{y}_F - \bar{y}_M) - 0}{se} = \frac{8.3 - 8.9}{1.09} = -0.55$$

(The difference we found is 0.55 se below the value under H_0, is this significant?)

Example

4. P-value

From Table B: $df = n_F + n_M - 2 = 486 + 354 - 2 = 838$

Our calculated |t| of 0.55 is smaller than $t_{.100}$, therefore p > 0.2

Alternatively, view -0.55 as z-score

From Table A: right-tail probability is 0.2912

$p = 2 * 0.2912 = 0.5825$

5. Conclusion

$p > \alpha$, do not reject H_0, no significant difference in number of close friends found between men and women

What about dependent samples?

- When samples are dependent, we need a **paired** t-test for 2 means
- We can now make matched pairs and calculate a difference score per pair
- Then do a t-test for a single mean on the difference scores!

Example paired t-test

- "Do both partners in the relationship have the same number of close friends?"

Partner 1	Partner 2
6	8
10	16
3	4
9	26
0	1
11	16

- Paired observations!

Example paired t-test

- "Do both partners in the relationship have the same number of close friends?"

Partner 1	Partner 2	Difference
6	8	2
16	10	-6
3	4	1
9	26	15
1	0	-1
11	16	5

- Use the difference scores and do a t-test about one mean

Introducing models

- A **model** is a (simple) approximation of relationships between variables in the population
  - E.g., to describe a normal distribution: $Y \sim N(\mu, \sigma)$
- Comparing means in two groups implies comparing two (normal) distributions
- Means concern μ, but what about the σs?

Equal or unequal standard deviations

Previous model:

$$H_0: Y_1 \sim N(\mu, \boxed{\sigma_1})$$
$$Y_2 \sim N(\mu, \boxed{\sigma_2})$$

$$H_a: Y_1 \sim N(\mu_1, \sigma_1)$$
$$Y_2 \sim N(\mu_2, \sigma_2)$$

Alternative model:

$$H_0: Y_1 \sim N(\mu, \boxed{\sigma})$$
$$Y_2 \sim N(\mu, \boxed{\sigma})$$

$$H_a: Y_1 \sim N(\mu_1, \sigma)$$
$$Y_2 \sim N(\mu_2, \sigma)$$

Assumptions, assumptions!

Unequal variances:

H_0: $Y_1 \sim N(\mu, \sigma_1)$

$Y_2 \sim N(\mu, \sigma_2)$

H_a: $Y_1 \sim N(\mu_1, \sigma_1)$

$Y_2 \sim N(\mu_2, \sigma_2)$

$$se = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Equal variances:

H_0: $Y_1 \sim N(\mu, \sigma)$

$Y_2 \sim N(\mu, \sigma)$

H_a: $Y_1 \sim N(\mu_1, \sigma)$

$Y_2 \sim N(\mu_2, \sigma)$

$$s = \sqrt{\frac{(n_1-1){s_1}^2+(n_2-1){s_2}^2}{n_1+n_2-2}}$$ **(pooled estimate)**

$$se = s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Effect size

- We can calculate statistical significance, but also **effect size**
- Remember: statistical significance is not practical significance!

- Effect size = $\frac{\bar{y}_1 - \bar{y}_2}{s}$, where s is again the pooled estimate (see previous slide)
- Effect is considered 'quite large' if |effect size| > 1

Comparing proportions in two groups

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What about proportions?

- Same principle as for means: can compare two groups using a confidence interval or a significance test

Death from Cancer			
Group	**Yes**	**No**	**Total**
Placebo	347	11,188	11,535
Aspirin	327	13,708	14,035

Confidence interval

A confidence interval for the difference $(\pi_2 - \pi_1)$ between two population proportions:

$$(\hat{\pi}_2 - \hat{\pi}_1) \pm z * se, \qquad se = \sqrt{\frac{\hat{\pi}_1(1-\hat{\pi}_1)}{n_1} + \frac{\hat{\pi}_2(1-\hat{\pi}_2)}{n_2}}$$

Requires:

- A categorical variable observed in two groups
- Independent random samples from the two groups
- Large enough samples s.t. there at least 10 successes and 10 failures in each group

https://istats.shinyapps.io/2sample_prop/

Significance test

Significance test for the difference between two population proportions:

$$z = \frac{(\hat{\pi}_2 - \hat{\pi}_1) - 0}{se_0}, \qquad \boxed{se_0} = \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n_1} + \frac{\hat{\pi}(1-\hat{\pi})}{n_2}}$$

where $\hat{\pi}$ is the pooled estimate of the two groups

Requires:

- A categorical variable observed in two groups
- Independent random samples from the two groups
- Large enough samples s.t. at least 5 successes and 5 failures in each group (two-sided)

https://istats.shinyapps.io/2sample_prop/

Bootstrap sampling

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When other methods fail

- Methods described so far have certain requirements
  - E.g., on normality of population distribution and/or sample size

- When these requirements are not met, you can use **bootstrap sampling**.

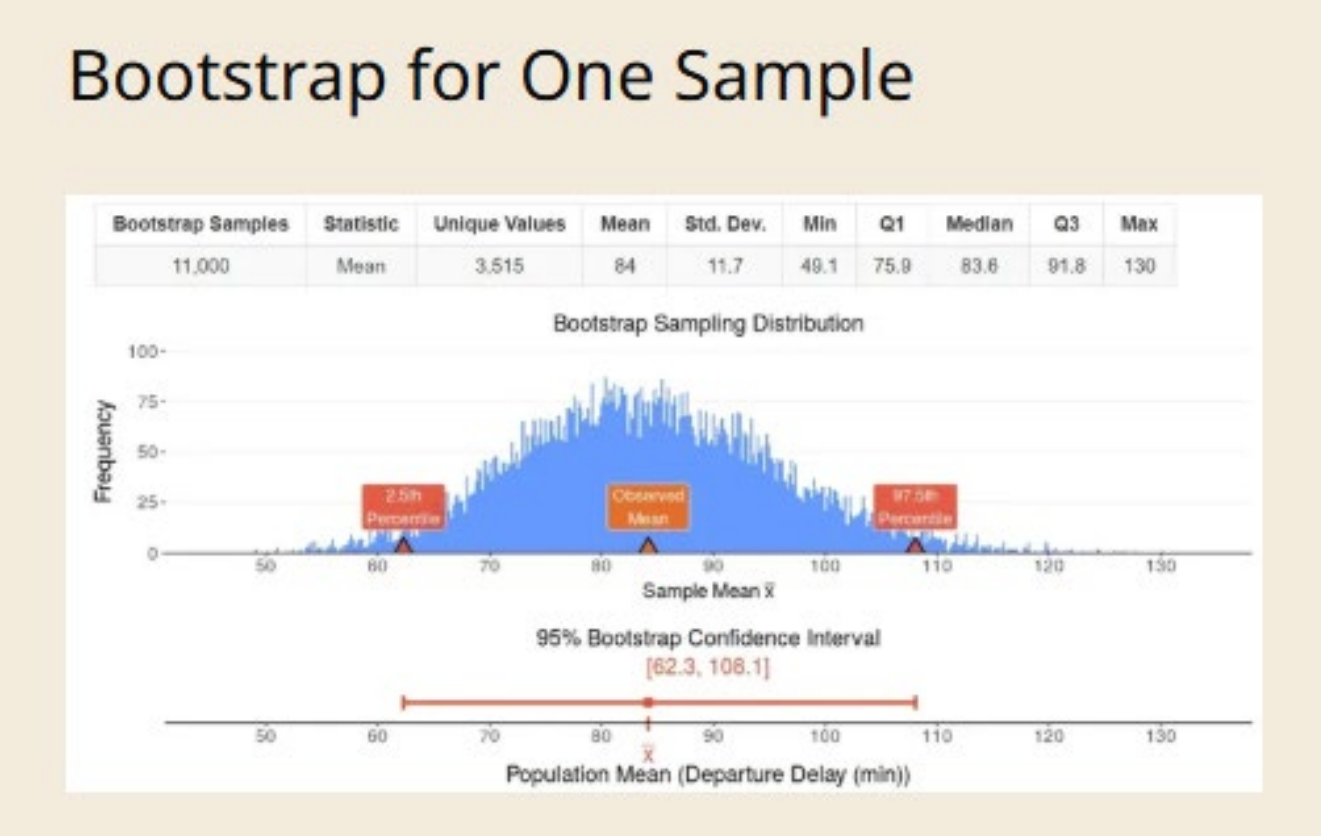
- **The bootstrap provides a way of approximating the sampling distribution of a statistic via simulation.**

Bootstrap sampling

- A **bootstrap sample** is sample drawn *with replacement* from the original sample and *of the same size* as the original sample.
- For a bootstrap sample, we can compute a statistic of interest.
- If we do this for many bootstrap samples, we obtain a **bootstrap distribution** that approximates the sampling distribution.

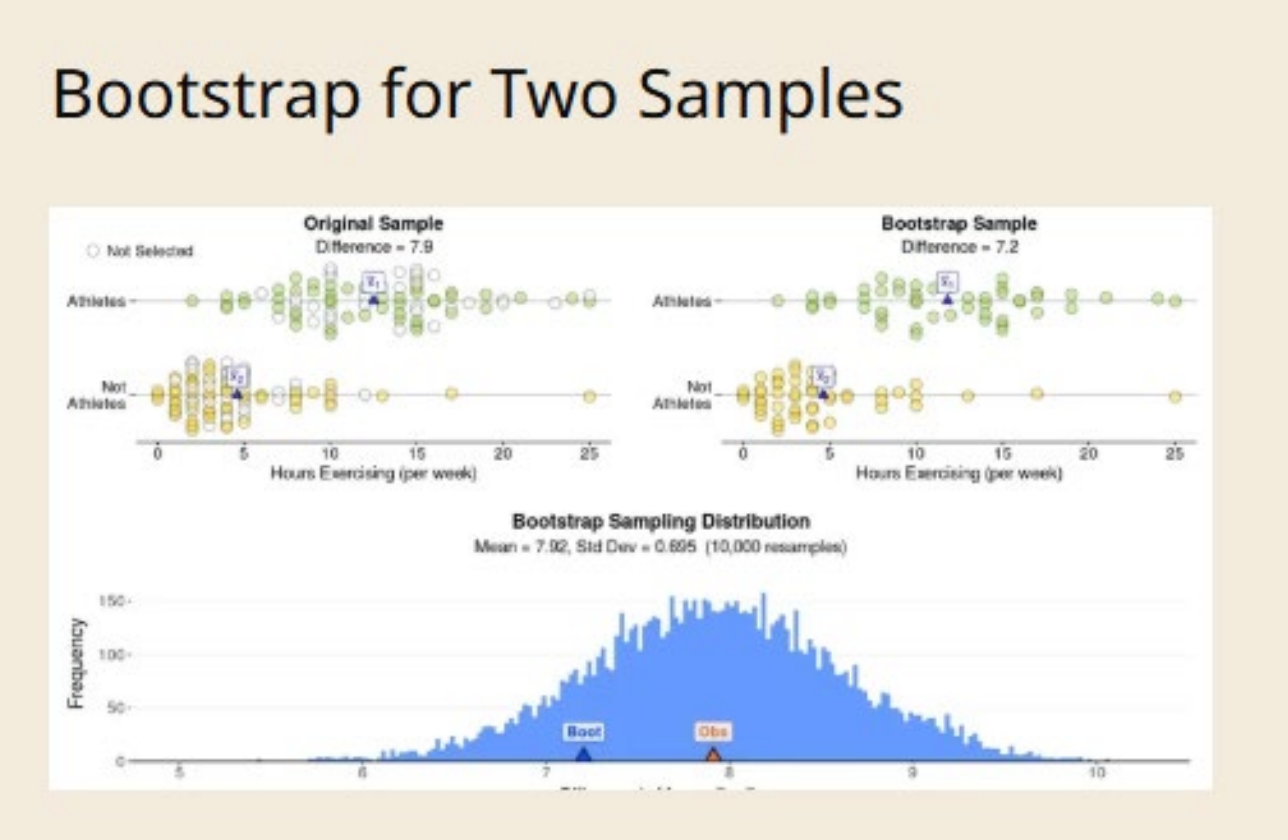
- Advantage: can be used for any statistic, population, and sample
- Disadvantage: cannot realistically be done 'by hand'

Bootstrap for one sample



https://istats.shinyapps.io/Boot1samp/

Bootstrap for two samples



https://istats.shinyapps.io/Boot2samp/

Wrapping up

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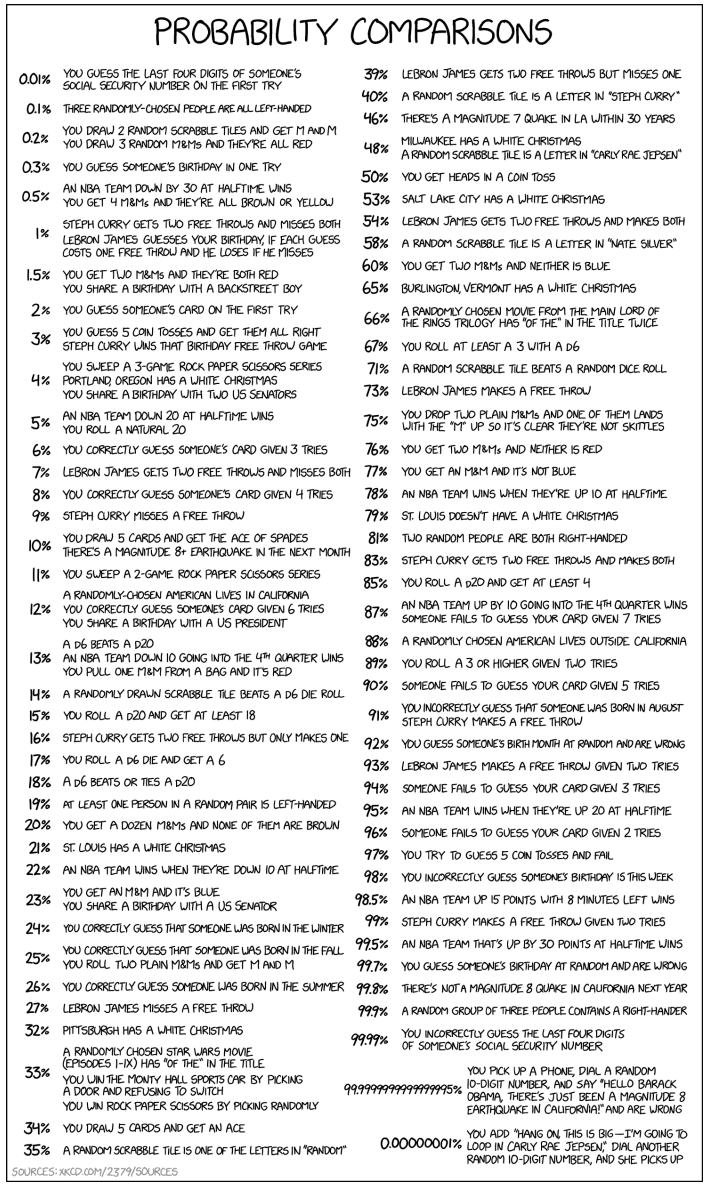
Wrapping up

- Read Chapter 10 of the book (excluding 10.5)
- Note that not all topics were covered in the lecture!

- Next: associations between two categorical variables



https://xkcd.com/2379/